

Week 5 — Lecture notes

Aperiodic signals and the Fourier transform

The figures below are snapshots of interactive widgets. The live versions, where you can move the sliders yourself, are at tchaffey.com/elec2302.

1. Quiz 1 review/questions
2. Periodic $\rightarrow$ aperiodic: motivation
3. Fourier transform as the limit of the Fourier series
4. Fourier transform vs Fourier series: what changed?
5. Fourier transform pairs
6. Properties of the Fourier transform.
7. Examples: radar, optics, RLC.

Last week: we represented a periodic signal as the sum of sinusoids, using the Fourier series.

This gave us a spectrum: the frequency content of the signal.

But: most signals are not periodic: a radar pulse, a plucked note, a spoken word ...

Question of the lecture: what does the spectrum of a one-off signal look like?

Core idea: we can take the Fourier series of a T -periodic signal and let $T \rightarrow \infty$ to get an aperiodic signal.

Periodic $\rightarrow$ aperiodic: motivation

Let's illustrate this by example: a pulse train, like a radar signal.

Take

$$u_T(t) = A \sum_{m \in \mathbb{Z}} p_\tau(t - mT), \quad p_\tau(t) = \begin{cases} 1, & |t| < \frac{\tau}{2} \\ 0, & |t| > \frac{\tau}{2} \end{cases}, \quad 0 < \tau < T.$$

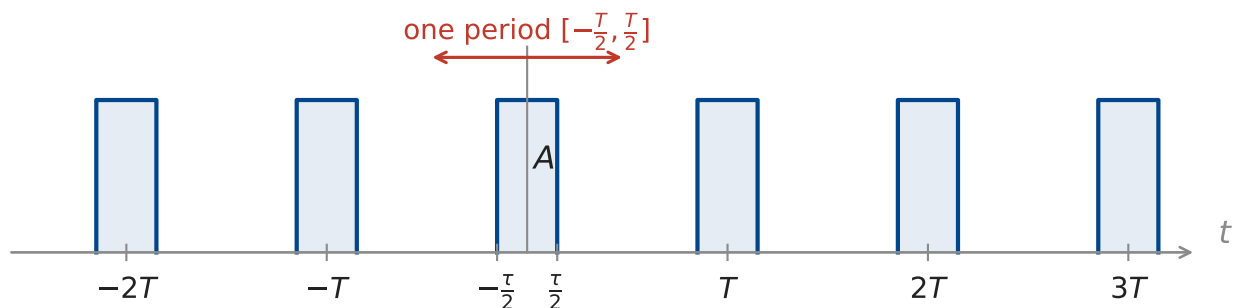


Figure 1: The pulse train $u_T(t)$: amplitude A , pulse width τ , period T . The red bracket marks the period $[-\frac{T}{2}, \frac{T}{2}]$ used for the coefficient integral.

τ varies on time. T varies the period. The ratio $\frac{\tau}{T}$ is called the “duty cycle”.

Let's calculate the Fourier series coefficients over the red period, $[-\frac{T}{2}, \frac{T}{2}]$. This contains a single pulse.

$$\begin{aligned} c_k &= \frac{1}{T} \int_{-T/2}^{T/2} u_T(t) e^{-jk\omega_0 t} dt \quad \left(\omega_0 = \frac{2\pi}{T} \right) \\ &= \frac{A}{T} \int_{-\tau/2}^{\tau/2} e^{-jk\omega_0 t} dt \quad \left(T = \frac{2\pi}{\omega_0} \right) \end{aligned}$$

$k = 0$:

$$c_0 = \frac{A}{T} \int_{-\tau/2}^{\tau/2} 1 dt = \frac{A\tau}{T}.$$

Use evenness — only need cos series.

$k \neq 0$:

$$\begin{aligned} c_k &= \frac{A}{T} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-\tau/2}^{\tau/2} = \frac{A\omega_0}{2\pi} \frac{1}{-jk\omega_0} (e^{-jk\omega_0 \frac{\tau}{2}} - e^{jk\omega_0 \frac{\tau}{2}}) \\ &= \frac{Aj}{2\pi k} \left(-2j \sin\left(\frac{k\omega_0 \tau}{2}\right) \right) \\ &= \frac{A}{\pi k} \sin\left(k \frac{\pi \tau}{T}\right) \\ &= A \frac{\sin(\pi k d)}{\pi k} \quad \left(d = \frac{\tau}{T} \right) \end{aligned}$$

$c_k = A d \operatorname{sinc}(kd).$

Recall: $\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$.

Fourier transform as the limit of the Fourier series

Now, how does this depend on the period T ?

$$\begin{aligned} c_k &= A \frac{\tau}{T} \operatorname{sinc}\left(k \frac{\tau}{T}\right) \\ &= A \frac{\tau}{T} \operatorname{sinc}\left(k \frac{\tau \omega_0}{2\pi}\right). \end{aligned}$$

Call $k\omega_0 = \omega$. Then

$$T c_k = A\tau \operatorname{sinc}\left(\frac{\tau \omega}{2\pi}\right) =: E(\omega) \quad (\text{“envelope”})$$

Where are the zeros?

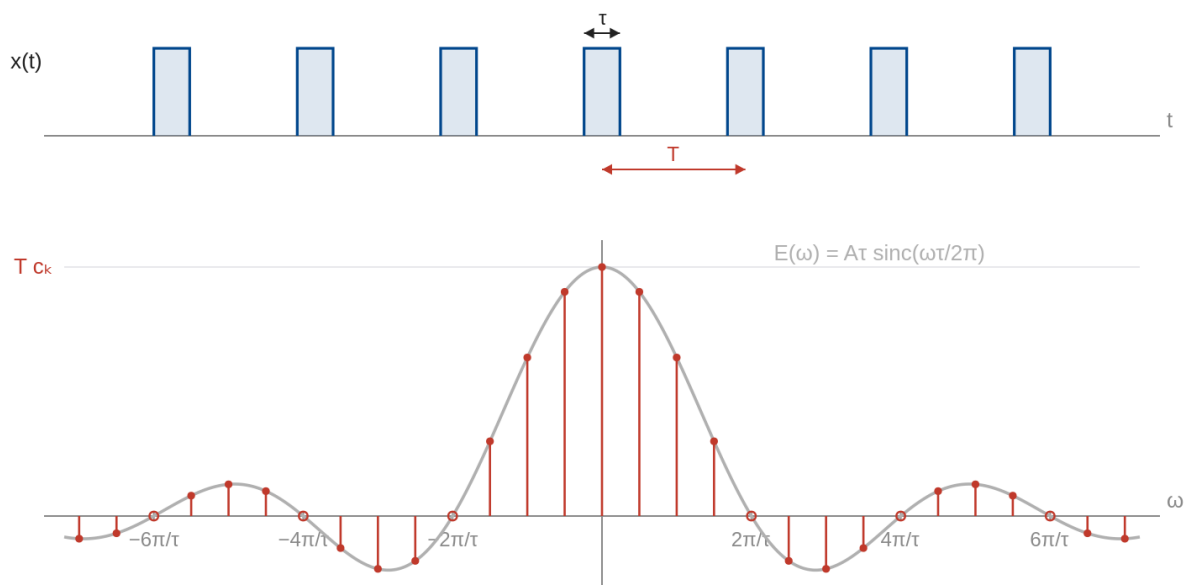


Figure 2: The pulse train (top) and its coefficients (bottom): the lines Tc_k (brick) sampling the fixed envelope $E(\omega) = A\tau \text{sinc}(\omega\tau/2\pi)$ (grey) at spacing $\omega_0 = 2\pi/T$. Shown at $T = 4\tau$, in the Tc_k view.

$$\text{sinc}\left(\frac{\tau\omega}{2\pi}\right) = 0$$

$$\sin\left(\frac{\tau\omega}{2}\right) = 0$$

$$\frac{\tau\omega}{2} = n\pi \quad \text{for } n \in \mathbb{Z}$$

$$\omega = \frac{n2\pi}{\tau}.$$

What happens as $T \rightarrow \infty$?

$E(\omega)$ doesn't change

Spacing becomes smaller.

What happens to the coefficients c_k ?

Let's define $u(t) = \lim_{T \rightarrow \infty} u_T(t)$.

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} u_T(t) e^{-jk\omega_0 t} dt$$

We define the Fourier transform of $u(t)$ by

$$U(j\omega) := \lim_{T \rightarrow \infty} Tc_k \Big|_{k\omega_0 \rightarrow \omega} = \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt.$$

This can be calculated for any signal for which the integral converges.

We know that the Fourier series can be used to reconstruct the original signal:

$$u_T = \sum_{k=-\infty}^{\infty} c_k \phi_k = \sum_{k=-\infty}^{\infty} \langle u_T, \phi_k \rangle_P \phi_k, \quad \phi_k(t) = e^{jk\omega_0 t}$$

Week 4 lecture.

Do we get something similar in the Fourier transform limit?

We can rewrite the Fourier series as

$$\begin{aligned} u_T(t) &= \sum_{k=-\infty}^{\infty} T c_k e^{jk\omega_0 t} \frac{1}{T} \quad [\text{trying to get a "d}\omega\text{" on the RHS}] \\ &= \sum_{k=-\infty}^{\infty} T c_k e^{jk\omega_0 t} \frac{\omega_0}{2\pi} \quad [\omega_0 = \text{spacing between samples}] \\ &= \sum_{k=-\infty}^{\infty} T c_k e^{j\omega t} \frac{\Delta\omega}{2\pi}. \end{aligned}$$

Then in the limit,

$$u(t) = \lim_{T \rightarrow \infty} \sum_{k=-\infty}^{\infty} T c_k e^{j\omega t} \frac{\Delta\omega}{2\pi}.$$

$$u(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega) e^{j\omega t} d\omega \quad (\text{the Inverse Fourier transform}).$$

We have illustrated this above for a pulse chain, but it holds for any signal for which the Fourier transform integral converges ($\int_{-\infty}^{\infty} |u(t)| dt < \infty$ is sufficient).

What has changed versus the Fourier series?

1. Sum $\mapsto$ integral, discrete index $k \mapsto$ continuous variable, $k\omega_0 \mapsto \omega$.
2. Units change. c_k has the same units as u . $U(j\omega)$ has units of $u \times \text{time}$: it is a density. Must be integrated to get anything.

Example: the Kurnell weather radar from the week 2 tutorial.

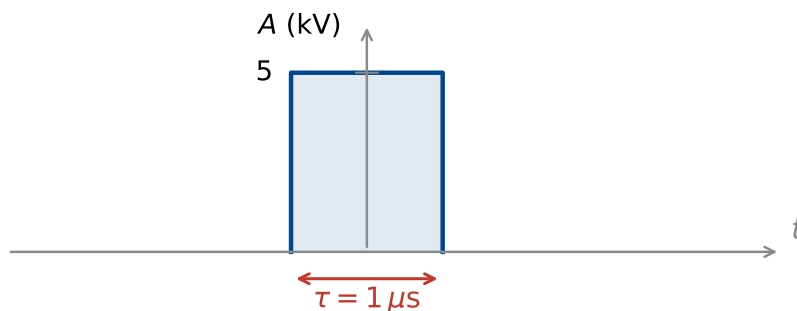


Figure 3: A single transmitted pulse: amplitude 5 kV, width $\tau = 1 \mu\text{s}$.

The parameters τ and T are varied independently:

τ adjusts range resolution:

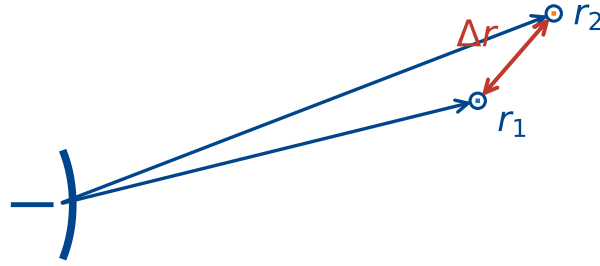


Figure 4: Two targets at ranges r_1 and r_2 , separated by Δr .

Consider two targets at ranges r_1 and r_2 , with $\Delta r := r_2 - r_1$.

The round trip time for each pulse $t_d = \frac{2r}{c}$ ($c = 3 \times 10^8 \text{ m s}^{-1}$).

$$\Delta t_d = \frac{2\Delta r}{c}.$$

If $\Delta t_d < \tau$, the returning pulses overlap and can't be distinguished.

The minimum distance that can be measured is therefore $\Delta r = \frac{c\tau}{2}$. For Kurnell, $\tau = 1 \mu\text{s}$, so $\Delta r \approx 150 \text{ m}$.

Why not make τ even smaller?

Let's check the spectrum again.

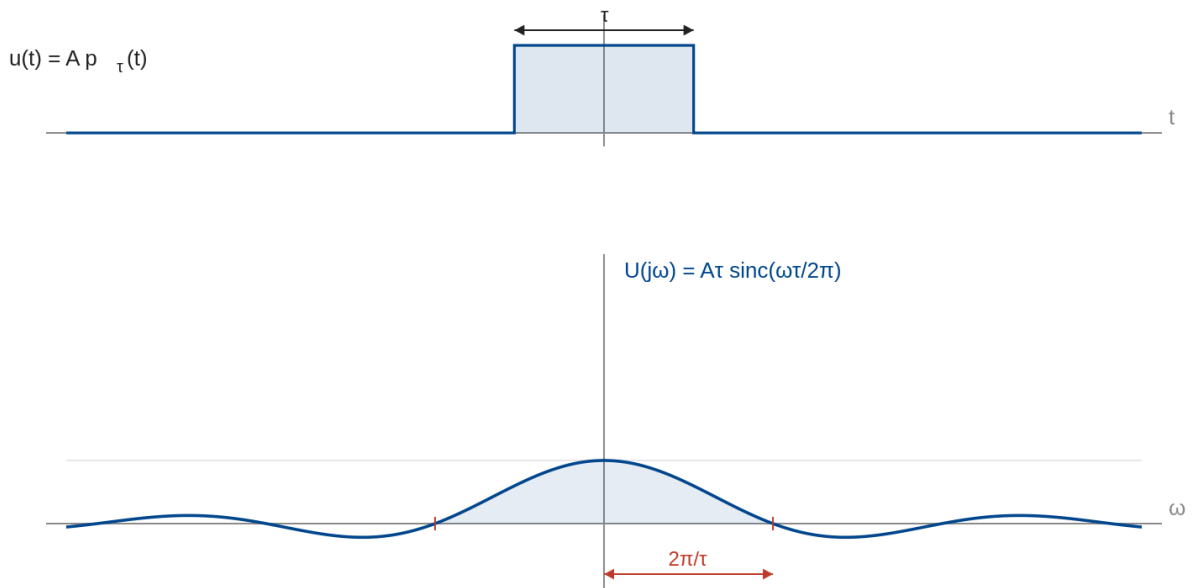


Figure 5: The width- τ pulse $u(t) = A p_\tau(t)$ (top) and its transform $U(j\omega) = A\tau \text{sinc}(\omega\tau/2\pi)$ (bottom), with the main lobe shaded and the first null at $2\pi/\tau$ marked. Shown at $\tau = 1$.

(Higher lobes are filtered — week 10.)

Small τ makes $\frac{2\pi}{\tau}$ big, so the lobe spreads and we need more bandwidth to transmit the pulse. Bandwidth is tightly regulated and very expensive!

Fourier transform pairs

The square $\leftrightarrow$ sinc Fourier transform pair is useful for radar design. Let's derive some other useful pairs.

(a) $u(t) = \delta(t)$.

$$\begin{aligned} U(j\omega) &= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt \\ &= 1 \quad (\text{sifting property}). \end{aligned}$$

(b) $u(t) = e^{-at}H(t)$

$$\begin{aligned} U(j\omega) &= \int_{-\infty}^{\infty} e^{-at}H(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(a+j\omega)t} dt \\ &= \left[\frac{-1}{a+j\omega} e^{-(a+j\omega)t} \right]_0^{\infty} \\ &= \frac{1}{a+j\omega}. \end{aligned}$$

Extension material

$u(t) = 1$

Skip this in lecture. Starred section in notes.

(c) $u(t) = 1$.

Treat this as the limit of a wide rectangle:

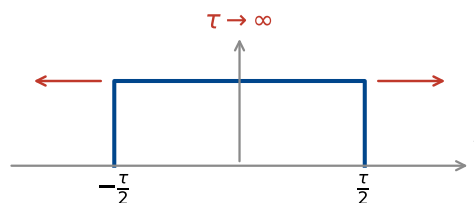


Figure 6: The rectangle widened to a constant.

We already have the Fourier transform of the rectangle: $\tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right)$.

What is the area under this curve?

$$\begin{aligned} A &= \int_{-\infty}^{\infty} \tau \frac{\sin\left(\frac{\tau\omega}{2}\right)}{\frac{\tau\omega}{2}} d\omega \\ &= \int_{-\infty}^{\infty} \frac{2 \sin\left(\frac{\tau\omega}{2}\right)}{\omega} d\omega. \end{aligned}$$

Let $\mu = \frac{\tau\omega}{2} \iff \omega = \frac{2\mu}{\tau}$.

$d\mu = \frac{\tau}{2} d\omega \iff d\omega = \frac{2 d\mu}{\tau}$.

$$\begin{aligned} A &= \int_{-\infty}^{\infty} \frac{2 \sin(\mu)}{\frac{2\mu}{\tau}} \frac{2 d\mu}{\tau} \\ &= \int_{-\infty}^{\infty} \frac{2 \sin(\mu)}{\mu} d\mu \\ &= \int_0^{\infty} \frac{4 \sin(\mu)}{\mu} d\mu \quad (\text{even function}). \end{aligned}$$

Fact: $\int_0^{\infty} \frac{\sin(\mu)}{\mu} d\mu = \frac{\pi}{2}$ (full derivation in the notes).

So the Fourier transform of the width τ pulse has constant area 2π .

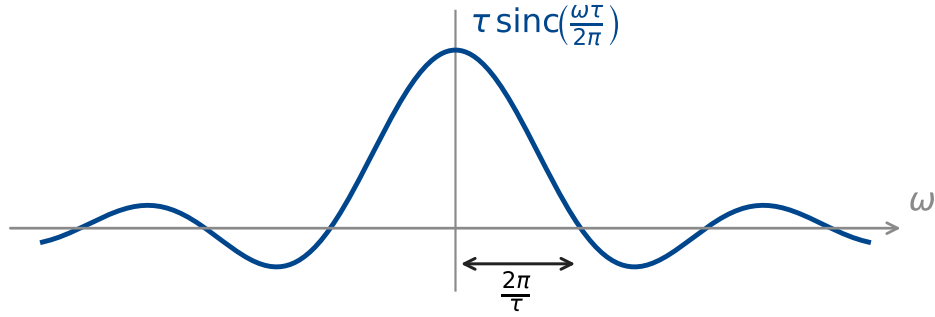


Figure 7: The transform of the width- τ pulse, with the main lobe half-width $\frac{2\pi}{\tau}$ marked.

As we increase τ , the sinc gets taller and narrower.

In the limit $\tau \rightarrow \infty$, we have an impulse with area 2π .

So $1 \xrightarrow{\text{FT}} 2\pi\delta(\omega)$.

(d) $e^{j\omega_0 t} \rightarrow 2\pi\delta(\omega - \omega_0)$

We already have the Fourier transform of 1:

$$\begin{aligned} \int_{-\infty}^{\infty} 1 \cdot e^{-j\omega t} dt &= 2\pi\delta(\omega). \\ \int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t} dt &= \int_{-\infty}^{\infty} e^{-j(\omega - \omega_0)t} dt = 2\pi\delta(\omega - \omega_0). \end{aligned}$$

So

$$\cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) \xrightarrow{\text{FT}} \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

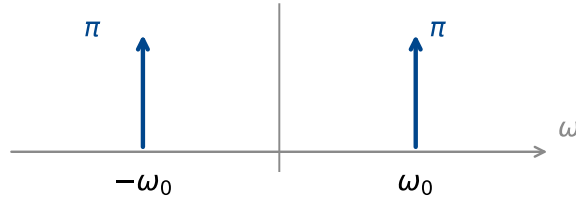


Figure 8: The spectrum of $\cos(\omega_0 t)$: two impulses of weight π at $\pm\omega_0$.

The Fourier series!

Properties of the Fourier transform

This is also a useful general property of the Fourier transform:

$$e^{j\omega_0 t} u(t) \leftrightarrow U(j(\omega - \omega_0)) \quad (\text{frequency shift})$$

There are several other useful properties:

$u(t - t_0)$:

$$\begin{aligned} \int_{-\infty}^{\infty} u(t - t_0) e^{-j\omega t} dt &= \int_{-\infty}^{\infty} u(\tau) e^{-j\omega(\tau + t_0)} d\tau \quad [\tau = t - t_0, \quad t = \tau + t_0, \quad d\tau = dt] \\ &= e^{-j\omega t_0} U(j\omega). \end{aligned}$$

$$u(t - t_0) \leftrightarrow e^{-j\omega t_0} U(j\omega) \quad (\text{time shift}).$$

$u(at)$:

$$\begin{aligned} &\int_{-\infty}^{\infty} u(at) e^{-j\omega t} dt. \quad \text{Let } \tau = at, \quad d\tau = a dt, \quad t = \frac{\tau}{a}. \\ &= \frac{a}{|a|} \int_{-\infty}^{\infty} u(\tau) e^{-j\omega \frac{\tau}{a}} \frac{d\tau}{a} \quad [\text{to get the limits the right way}] \\ &= \frac{1}{|a|} \int_{-\infty}^{\infty} u(\tau) e^{-j\frac{\omega}{a} \tau} d\tau \\ &= \frac{1}{|a|} U\left(\frac{j\omega}{a}\right). \end{aligned}$$

$$u(at) \leftrightarrow \frac{1}{|a|} U\left(\frac{j\omega}{a}\right) \quad (\text{time scaling}).$$

$$\alpha u_1(t) + \beta u_2(t) \leftrightarrow \alpha U_1(j\omega) + \beta U_2(j\omega) \quad (\text{linearity}).$$

$y = h * u$.

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(t - \tau) u(\tau) d\tau \\ Y(j\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(t - \tau) u(\tau) d\tau e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(t - \tau) e^{-j\omega t} dt u(\tau) d\tau \quad [\text{using Fubini's theorem}] \end{aligned}$$

Let $\mu = t - \tau$. $dt = d\mu$. $t = \mu + \tau$.

$$\begin{aligned} Y(j\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\mu) e^{-j\omega(\mu+\tau)} d\mu u(\tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\mu) e^{-j\omega\mu} d\mu \int_{-\infty}^{\infty} u(\tau) e^{-j\omega\tau} d\tau \\ Y(j\omega) &= H(j\omega) U(j\omega) \end{aligned}$$

$$\boxed{y = h * u \leftrightarrow Y = HU} \quad (\text{convolution})$$

This is the foundation of the rest of the course.

$$\frac{d}{dt}u(t) \rightsquigarrow \int_{-\infty}^{\infty} \frac{d}{dt}u(t) e^{-j\omega t} dt$$

Integrate by parts: $\int u'v = uv - \int uv'$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{d}{dt}u(t) e^{-j\omega t} dt &= \underbrace{[u(t)e^{-j\omega t}]_{-\infty}^{\infty}}_{=0 \text{ if } u(-\infty)=u(\infty)=0} + j\omega \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt \\ &= j\omega U(j\omega). \end{aligned}$$

So

$$\boxed{\frac{d}{dt}u(t) \leftrightarrow j\omega U(j\omega).}$$

Summary

Transform pairs.

$u(t)$	$U(j\omega)$	derived in
$A p_{\tau}(t)$	$A\tau \operatorname{sinc}\left(\frac{\omega\tau}{2\pi}\right)$	§2
$\delta(t)$	1	(a)
$e^{-at}H(t)$	$\frac{1}{a + j\omega}$	(b)
1	$2\pi\delta(\omega)$	(c)
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	(d)
$\cos(\omega_0 t)$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$	(d)

Properties.

	$u(t)$	$U(j\omega)$
frequency shift	$e^{j\omega_0 t}u(t)$	$U(j(\omega - \omega_0))$
time shift	$u(t - t_0)$	$e^{-j\omega t_0}U(j\omega)$
time scaling	$u(at)$	$\frac{1}{ a }U\left(\frac{j\omega}{a}\right)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(j\omega) + \beta U_2(j\omega)$
convolution	$y = h * u$	$Y = HU$
differentiation	$\frac{d}{dt}u(t)$	$j\omega U(j\omega)$
duality	$X(jt)$	$2\pi x(-\omega)$

	$u(t)$	$U(j\omega)$
Parseval	$\int_{-\infty}^{\infty} u(t) \overline{y(t)} dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega) \overline{Y(j\omega)} d\omega$

Other properties

(Proofs in lecture notes.)

Duality: if $x(t) \rightsquigarrow X(j\omega)$, then

$$X(jt) \rightsquigarrow 2\pi x(-\omega).$$

Proof:

$$\begin{aligned} x(-\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(js) e^{-j\omega s} ds \\ &= \frac{1}{2\pi} \mathcal{F}\{X(jt)\}(\omega). \end{aligned}$$

Example: we know $\delta(t) \rightsquigarrow 1$.

The dual gives us another pair: $1 \rightsquigarrow 2\pi\delta(\omega)$.

Parseval's theorem:

$$\int_{-\infty}^{\infty} u(t) \overline{y(t)} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega) \overline{Y(j\omega)} d\omega.$$

Special case: if $y = u$:

$$\int_{-\infty}^{\infty} |u(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |U(j\omega)|^2 d\omega.$$

Proof:

$$\begin{aligned} \int u \bar{y} dt &= \int u(t) \frac{1}{2\pi} \int \overline{Y(j\omega)} e^{j\omega t} d\omega dt \\ &= \frac{1}{2\pi} \int \left[\int u(t) e^{-j\omega t} dt \right] \overline{Y(j\omega)} d\omega \\ &= \frac{1}{2\pi} \int U(j\omega) \overline{Y(j\omega)} d\omega. \end{aligned}$$

Example: the RLC circuit

Example: RLC circuit with v_C as output.

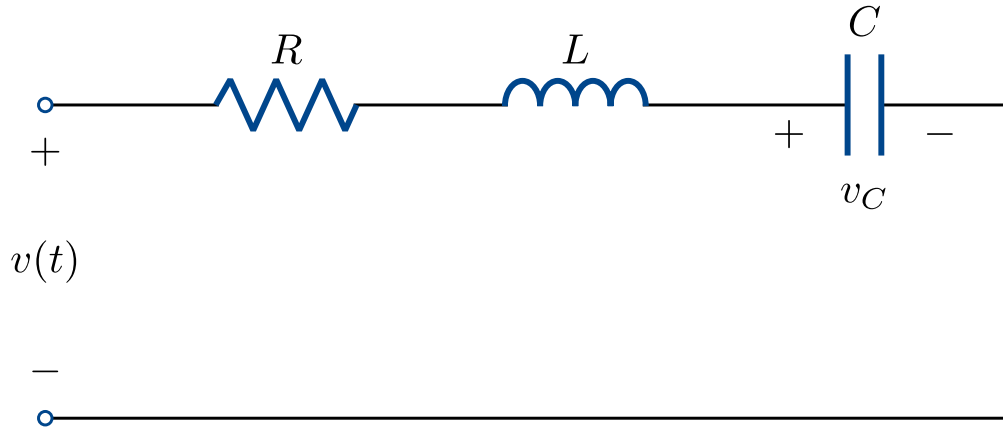


Figure 9: Series RLC circuit with v_C as the output (drawn with circuitikz).

We know the impulse response is

$$g(t) = 6.25 e^{-3t} \sin(4t) H(t).$$

What is the spectrum of this signal?

$$\sin(4t) = \frac{1}{2j} (e^{j4t} - e^{-j4t})$$

So

$$g(t) = \frac{6.25}{2j} (e^{(-3+4j)t} - e^{(-3-4j)t}).$$

Use the Fourier transform pair $e^{-at}H(t) = \frac{1}{a + j\omega}$:

$$\begin{aligned} G(j\omega) &= -3.125j \left(\frac{1}{-(-3+4j) + j\omega} - \frac{1}{-(-3-4j) + j\omega} \right) \\ &= -3.125j \left(\frac{(+3+4j+j\omega) - (+3-4j+j\omega)}{(+3-4j+j\omega)(+3+4j+j\omega)} \right) \\ &= -3.125j \left(\frac{8j}{(+3-4j)(+3+4j) + j\omega(+3-4j) + (+3+4j)j\omega - \omega^2} \right) \\ &= \frac{25}{-\omega^2 + 6j\omega + 25}. \end{aligned}$$

What does the spectrum look like?

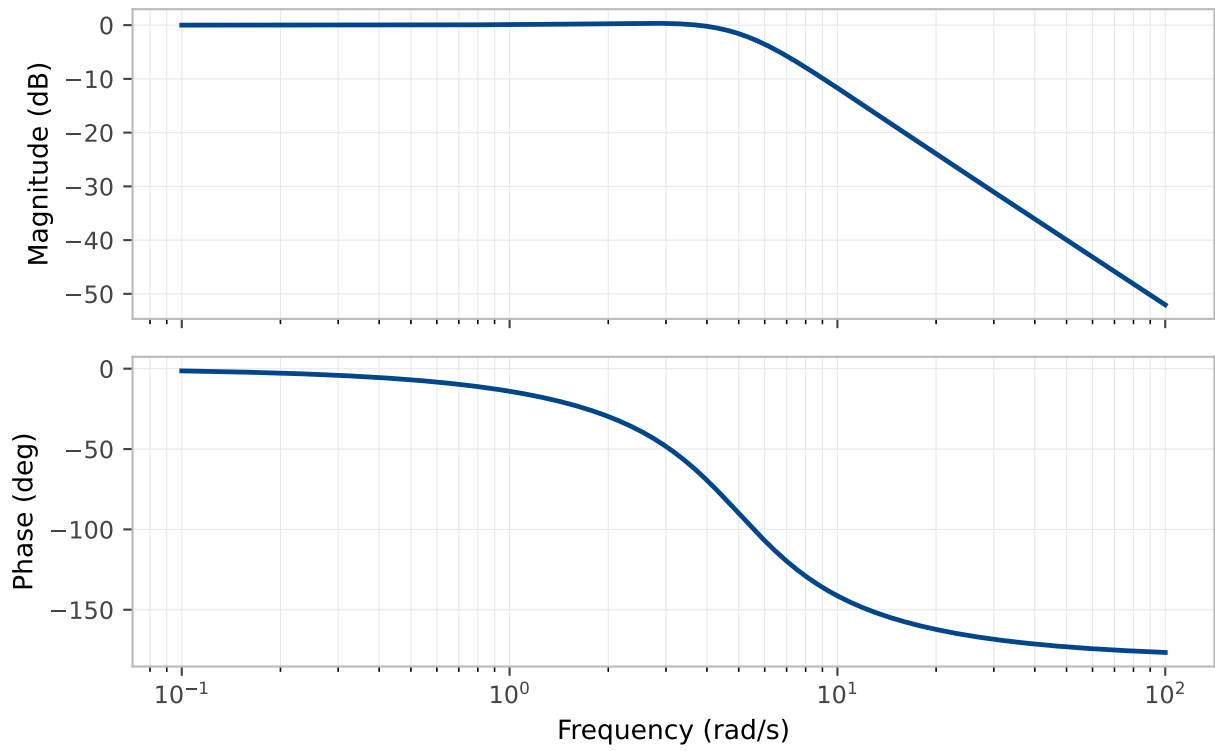


Figure 10: Magnitude and phase of $G(j\omega) = \frac{25}{-\omega^2 + 6j\omega + 25}$.